

Technical Document #366: Data Engineering - Comprehensive Overview

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ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Key Takeaways:

- Data governance establishes policies and procedures for managing data assets.
- Data quality ensures data is accurate, complete, and consistent.
- Data pipelines automate the movement and transformation of data between systems.